MindFrame: The Whole Model as One Equation

The master function, the Kai shield, and a faith reading

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MindFrame is a model built from design assumptions and informed by research. It is not validated science, not a diagnosis, and not therapy. In a crisis in the US, call or text 988, or call 911.

All credit and glory to God. I owe my knowledge to my Lord Jesus Christ.

1 The whole model on one page

MindFrame reads one thought into a vector $e \in [0,1]^9$ of nine movers (fear, sadness, anger, shame, calm, connection, hope, rumination, avoidance) and takes one gentle step toward a home state h. Every feature built so far fits in four lines:

$$e = \mathrm{clip}_{[0,1]}\big(e_- \oplus n(\text{text}) + X\,(1-S)\,w\big) \qquad \text{the reading} \tag{1}$$

$$e' = \mathrm{clip}_{[0,1]}\big(e + a(e)\,(h-e) + b\,s(e)\big) \qquad \text{the step} \tag{2}$$

$$P = P_d(e)\,(1-\gamma C)\,(1-\delta G);\quad \ell = \arg\max_i \Delta P_i;\quad \text{step} = \text{practice}(\ell) \qquad \text{peace} \tag{3}$$

$$\text{words} = \text{compose}(e, e', P, \ell, \text{passes}). \tag{4}$$

- e_- is where the person was: the session's e, or the vector picked with the feeling chips. $n(\text{text})$ is what the words add: a lexicon, thinking-trap signals, and word hooks (relationship patterns, forgiveness cues, true size, patience, body tension), each saturating as $1 - e^{-\text{raw}}$. The saturating sum $\oplus$ is $1-(1-e_j)(1-n_j)$ for hard and process movers and $e_j + n_j^+(1-e_j) - n_j^- e_j$ for the grounding movers (calm, connection, hope).
- $X(1-S)w$ is the Kai term (Section 2). For almost every text $X = 0$, and line (1) is just $e_- \oplus n$.
- $a(e) = \min(a_{\max}, a_0 + k\|e-h\|)$, lowered to $a_{\text{cap}} = 0.12$ (with $b = b_U$) when the active hard movers feed each other ($\lambda > 0$). The term $s_j(e) = e_j \sum_g \min(0, W_{gj})\,h_g \le 0$ is grounding pressing on a hard or process mover that sits above home.
- $P_d = 1/(1+(d/d_0)^2)$ with $d = \|e-h\|$. C is the strongest hidden conflict $\sqrt{x_i y_j}$. $G = \max(g,u)$ is the true-size signal (too big, g; too small, u).
- ΔP_i is the gain in P when e_i moves 0.05 toward h_i. The lever ℓ is the largest gain. practice(ℓ) is a fixed, deterministic rule table: for example, the cold-water reset only for very high fear or anger with low peace, and the brain dump for very high rumination. Strictly, the step is the practice for the lever that maximizes ΔP; the model does not search every practice.
- The *passes* run in the same call and never move e. They are forgiveness ($F_{\text{self}} = e_{\text{shame}}(0.4 + 0.6c_{\text{self}})$ and $F_{\text{other}} = e_{\text{anger}}(0.4 + 0.6c_{\text{other}})$), the freeing mix ($\max J = P - \mu N$ over small band moves), the Kai lines, and the companion. The companion grows only from caring actions (an Update, a step tried, a journal save, a practice finished), never from e.
- Values are rounded to 0.01 after (1) and after (2), as the code does. The baseline $B_t = 0.8B_{t-1} + 0.2e_t$ and the recovery count (repeat (2) until $d < 0.3$) reuse the same map.

Code and formula agree. A unit test recomputes (1)–(3) from these lines alone and checks that `master()` returns exactly the same e, e', P, lever and practice. The refactor changed no other behavior: all earlier tests still pass.

2 The Kai equation: steadiness under outside pressure

In Ryan's story, Kai faces a storm of criticism. He anchors in self-awareness, applies forbearance and empathy, and his respect guard blocks the barbs without harming anyone, so impulsive reaction "flickers and fails." The model adds no movers:

$X = 1 - e^{-0.7k_X}$	hurt received (aimed *at* you)
$A = (1-G)\,[1-(1-c_A)(1-\frac{1}{2}\sigma_A)]$	self-awareness
$F = 1-(1-c_F)(1-\frac{1}{2}\sigma_F)(1-p_F)$	forbearance
$E = 1-(1-c_E)(1-\frac{1}{2}\sigma_E)$	empathy
$R = (1-c_{\text{attack}})\,[1-(1-c_R)(1-\frac{1}{2}\sigma_R)]$	respect guard
$S = \min\big(0.99,\ 1-(1-A)(1-F)(1-E)(1-R)\big)$	the shield
$e_j \leftarrow \mathrm{clip}_{[0,1]}\big(e_j + X(1-S)\,w_j\big),\quad w = (0.8, 0.6, 0.4)$ for anger, shame, fear	the push
$I = X(1-S)\,u,\quad u = \max(e_{\text{anger}}, c_{\text{urge}})$	impulsive reaction

Here each $c = 1 - e^{-0.7k}$ counts k matching phrases, with negation handled ("nobody judged me" gives $X = 0$). $\sigma = 1$ when a matching practice was finished this session: the pause or breathing practices for F, labeling for A, and the REACH "Empathize" step for E. Sessions are kept in memory only. $p_F = \frac{1}{2}$ comes from a picker chip. Each virtue is capped at 0.95. The Four Horsemen (Gottman and Levenson, 1992) in the person's *own* words never raise X; contempt or criticism aimed outward lowers R through c_{attack}.

On the card (words only, and only when $X > 0$), for the reading "My coworkers criticized me in front of everyone, I want to snap back but I know they're stressed": $X = 0.75$, $A = 0$, $F = 0.5$, $E = 0.5$, $R = 0$, $S = 0.75$, $I = 0.14$. The card says: "Your steadiness: self-awareness just starting, forbearance growing, empathy growing, respect guard just starting. The urge to react flickers and fails." The weakest virtue is the lever, and it comes with one step. For the respect guard, that step is DEAR MAN-style (Linehan, 2015): "I hear you. I see it differently, and I'm going to step away now."

Sources, honestly. Naming feelings dampens amygdala response (Lieberman et al., 2007); affirming core values buffers threat to the self (Cohen and Sherman, 2014). Patience relates to well-being and can be trained (Schnitker, 2012). Perspective taking is a distinct facet of empathy (Davis, 1983). Assertiveness training has a substantial, if older, evidence base (Speed et al., 2018). None of these studies tested this equation. The product form of S, the weights w, and the $\frac{1}{2}$ terms are modeling choices.

3 The faith reading

With *Include faith* on, the home h is described as rest and peace in Christ: "Come to me, all you who labor and are heavily burdened, and I will give you rest" (Matthew 11:28–30), and "the peace of God, which surpasses all understanding, will guard your hearts and your thoughts in Christ Jesus" (Philippians 4:7). The virtues are read as fruits of the Spirit: "love, joy, peace, patience, kindness, goodness, faith, gentleness, and self-control" (Galatians 5:22–23). Verse wording follows the World English Bible (public domain).

Model	Fruit of the Spirit	Picture in the Gospel
Forbearance F	patience	He "answered nothing" before his accusers (Matthew 27:12–14); "as a sheep that before its shearers is silent" (Isaiah 53:7)
Empathy E	kindness	"Father, forgive them, for they don't know what they are doing" (Luke 23:34)
Respect guard R	gentleness	"When he was cursed, he didn't curse back" (1 Peter 2:23)
Low impulsive reaction I	self-control	the same verse: "When he suffered, he didn't threaten"
Home h	peace	Philippians 4:7; Matthew 11:28–30

Only the words change, not the math. With faith on, every number (e, e', X, S, I, P) is identical; this is tested. The faith reading is offered as a way of speaking about the same quantities, not as a claim that the model measures anything spiritual.

4 Where ideas come from: one attribution equation

Every line of the one-page model (Section 1) stands on someone's research. Here the model is split into 18 terms $\tau_1, \ldots, \tau_{18}$, and each term is tagged with the credited sources that ground it. A source is tagged only where the code comments, the README mover table or the papers name it for that term, and a unit test checks that each tagged surname appears in the evidence file. Two terms, τ_5 and τ_6, are design choices with no single source, and they are left untagged. Colors group the terms: reading and words, the Kai shield, peace, lever and practice, the

passes, faith and story, design choices.

$$\mathcal{M} = \text{words}_{\tau_{18}}\Big(e' = \text{clip}\big(e + \underbrace{a(e)(h-e)}_{\tau_5} + \underbrace{b\,s(e)}_{\tau_6}\big),\quad e = \text{clip}\big(\underbrace{e_-}_{\tau_1} \oplus \underbrace{n(\text{text})}_{\tau_2} + \underbrace{X}_{\tau_3}(1 - \underbrace{S}_{\tau_4})\,w\big),$$

$$P = \underbrace{P_d}_{\tau_7}(1 - \gamma\,\underbrace{C}_{\tau_8})(1 - \delta\,\underbrace{G}_{\tau_9}),\quad \underbrace{\ell}_{\tau_{10}} \mapsto \underbrace{\text{practice}(\ell)}_{\tau_{11}},\quad \underbrace{F}_{\tau_{12}}, \underbrace{J}_{\tau_{13}}, \underbrace{x(1-L)}_{\tau_{14}}, \underbrace{\Pi}_{\tau_{15}}, \underbrace{\sigma}_{\tau_{16}}, \underbrace{\text{story}}_{\tau_{17}}\Big),$$

$$\tau_k \leftarrow \sum_{s \in \text{Src}(\tau_k)} s, \qquad \text{share}(s) = \frac{|\{k :\ s \in \text{Src}(\tau_k)\}|}{18}.$$

Here Π is past to future, σ the scripture seed (faith on only), and *story* the films and songs, credited to their studios and artists. The sum $\sum s$ means the ideas a term draws on. It is not arithmetic on the sources.

Term	Sources that ground it
τ_1 e The nine movers	Barlow et al. 2004; Beck et al. 1985; Kim et al. 2011; Zaccaro et al. 2018; Holt-Lunstad et al. 2010; Snyder 2002; Aldao et al. 2010; Nolen-Hoeksema et al. 2008; Bond et al. 2011; Gillanders et al. 2014
τ_2 $n(\text{text})$ Reading your words	Beck 1976; Burns 1980; Al-Mosaiwi & Johnstone 2018; Gottman 1994; Gottman & Levenson 1992; Gottman & Levenson 2000; Heyman & Slep 2001
τ_3 X Outside hurt	Gottman & Levenson 1992
τ_4 S The Kai shield	Lieberman et al. 2007; Cohen & Sherman 2014; Schnitker 2012; Davis 1983; Linehan 2015; Speed et al. 2018
τ_5 $a(e)(h-e)$ One gentle step home	*design choice, no single source*
τ_6 $b\,s(e)$ Feelings feeding each other	*design choice, no single source*
τ_7 P_d Peace near home	Desbordes et al. 2015; Kabat-Zinn 2003
τ_8 C Hidden conflicts	Lee et al. 2013; Miller & Rollnick 2013
τ_9 G True size	Beck 1976; Burns 1980; Al-Mosaiwi & Johnstone 2018; Tangney 2000; Neff 2003; Gilbert & Procter 2006; Gilbert 2009; Gross 1998
τ_{10} ℓ The lever	Gross 1998
τ_{11} $\text{practice}(\ell)$ The practice	Stahl & Goldstein 2010; Segal et al. 2013; Kabat-Zinn 1990; Lieberman et al. 2007; Neff 2003; Zaccaro et al. 2018; Gross 1998; Schnitker 2012; Bowen & Marlatt 2009; Emmons & McCullough 2003; Linehan 2015; Jacobson 1938; McCallie et al. 2006; Pennebaker & Beall 1986; Frattaroli 2006; Rebar et al. 2015; Ekkekakis et al. 2000
τ_{12} F Forgiveness	Cornish & Wade 2015; Neff 2003; Worthington 2006; Wade et al. 2014; Enright & Fitzgibbons 2015; McCullough et al. 2007
τ_{13} J The freeing mix	Tangney 2000; Burns 1980; Neff 2003; Gilbert & Procter 2006; Gilbert 2009; Teresa of Ávila
τ_{14} $x(1-L)$ Loosen the grip	Wegner et al. 1987; Bowen & Marlatt 2009; Borkovec et al. 1983; McGowan & Behar 2013; Masuda et al. 2004
τ_{15} Π Past to future	Neff 2003; Cornish & Wade 2015; Hofmann et al. 2011; Hayes et al. 2006; King 2001; Peters et al. 2010; Hershfield 2011
τ_{16} σ Scripture seed	World English Bible
τ_{17} story Continue your story	Zillmann 1988; Wikidata; TMDB
τ_{18} words The words you read	Burns 1980; Miller & Rollnick 2013; Alderson-Day & Fernyhough 2015; Fernyhough 2016; Vygotsky 1986; Kross et al. 2014; Cowan 2001; Miller 1956; Sweller 1988; Nielsen 2006

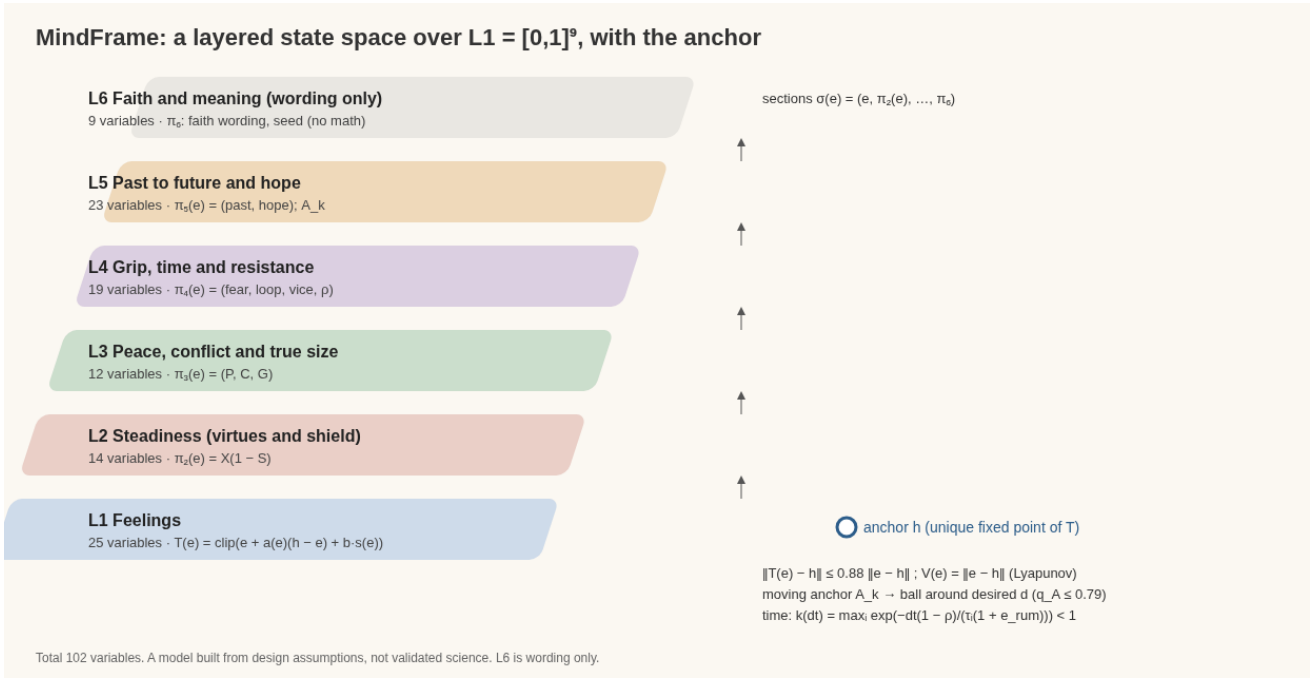
Contribution map (structural map, not effect size or validated weight). For each source, the terms it informs and its *structural share*, $\text{share}(s) = \text{terms informed} / 18$. The share counts where an idea appears in the model's structure. It is not an effect size, not a validated weight, and not a ranking of how much any researcher's work matters. No effect sizes are estimated here. 70 of the 101 credited sources inform at least one term:

Source	Terms	Share	Source	Terms	Share
Neff 2003	9,11,12,13,15	5/18	Hershfield 2011	15	1/18
Burns 1980	2,9,13,18	4/18	Heyman & Slep 2001	2	1/18
Gross 1998	9,10,11	3/18	Hofmann et al. 2011	15	1/18
Al-Mosaiwi & Johnstone 2018	2,9	2/18	Holt-Lunstad et al. 2010	1	1/18
Beck 1976	2,9	2/18	Jacobson 1938	11	1/18
Bowen & Marlatt 2009	11,14	2/18	Kabat-Zinn 1990	11	1/18
Cornish & Wade 2015	12,15	2/18	Kabat-Zinn 2003	7	1/18
Gilbert & Procter 2006	9,13	2/18	Kim et al. 2011	1	1/18
Gilbert 2009	9,13	2/18	King 2001	15	1/18
Gottman & Levenson 1992	2,3	2/18	Kross et al. 2014	18	1/18
Lieberman et al. 2007	4,11	2/18	Lee et al. 2013	8	1/18
Linehan 2015	4,11	2/18	Masuda et al. 2004	14	1/18
Miller & Rollnick 2013	8,18	2/18	McCallie et al. 2006	11	1/18
Schnitker 2012	4,11	2/18	McCullough et al. 2007	12	1/18
Tangney 2000	9,13	2/18	McGowan & Behar 2013	14	1/18
Zaccaro et al. 2018	1,11	2/18	Miller 1956	18	1/18
Aldao et al. 2010	1	1/18	Nielsen 2006	18	1/18
Alderson-Day & Fernyhough 2015	18	1/18	Nolen-Hoeksema et al. 2008	1	1/18
Barlow et al. 2004	1	1/18	Pennebaker & Beall 1986	11	1/18
Beck et al. 1985	1	1/18	Peters et al. 2010	15	1/18
Bond et al. 2011	1	1/18	Rebar et al. 2015	11	1/18
Borkovec et al. 1983	14	1/18	Segal et al. 2013	11	1/18
Cohen & Sherman 2014	4	1/18	Snyder 2002	1	1/18
Cowan 2001	18	1/18	Speed et al. 2018	4	1/18
Davis 1983	4	1/18	Stahl & Goldstein 2010	11	1/18
Desbordes et al. 2015	7	1/18	Sweller 1988	18	1/18
Ekkekakis et al. 2000	11	1/18	Teresa of Ávila	13	1/18
Emmons & McCullough 2003	11	1/18	TMDB	17	1/18
Enright & Fitzgibbons 2015	12	1/18	Vygotsky 1986	18	1/18
Fernyhough 2016	18	1/18	Wade et al. 2014	12	1/18
Frattaroli 2006	11	1/18	Wegner et al. 1987	14	1/18
Gillanders et al. 2014	1	1/18	Wikidata	17	1/18
Gottman & Levenson 2000	2	1/18	World English Bible	16	1/18
Gottman 1994	2	1/18	Worthington 2006	12	1/18
Hayes et al. 2006	15	1/18	Zillmann 1988	17	1/18

The other 31 credited sources give context, ways to test the model, mathematics, or philosophy, and inform no term: Beck et al. 1979; Beck 2011; Hofmann et al. 2012; Wolpe 1969; Rachman & de Silva 1978; Holmes & Mathews 2010; Oettingen 2014; American Psychiatric Association. 2013; Kroenke et al. 2009; Shiffman et al. 2008; Dawes 1979; Kravitz & Duan 2014; Fitzpatrick et al. 2017; Linehan 1993; Neacsiu et al. 2010; Neacsiu et al. 2010; Hayes et al. 2012; Baumeister & Leary 1995; Meng et al. 2024; Whorf 1956; Boroditsky 2001; Weizenbaum 1966; Bommasani et al. 2021; Langlands 1967; Gelbart 1984; Wiles 1995; Frenkel 2013; Frankl 2006; Robertson 2010; Epictetus. (c. 125 CE). Enchiridion (Handbook). Various translations. undefined; Marcus Aurelius. (c. 170–180 CE). Meditations. Various translations. undefined.

5 A layered topology and the anchor

Layers. Every variable in the MindFrame equations (102 in all, listed in `js/core/topology.js`) sits on one of six layers: L1 feelings and the step (25), L2 virtues and the Kai shield (14), L3 peace, conflict and true size (12), L4 grip, time and resistance (19), L5 past to future, hope, forgiveness, the freeing mix and the moving anchor (23), L6 faith and meaning (9, *wording only*: the seed and the verse choice are not measured quantities). The state space is the product $E = L_1 \times \cdots \times L_6$ over the base $L_1 = [0,1]^9$. The upper layers are read from L_1 and the words by projection maps $\pi_2(e) = X(1-S)$, $\pi_3(e) = (P, C, G)$, $\pi_4(e) = (\text{fear}, \text{loop}, \text{vice}, \rho)$, $\pi_5(e) = (\text{past}, \text{hope})$, so a point over e is the section $\sigma(e) = (e, \pi_2(e), \ldots, \pi_6)$. We call this "fiber-bundle style" only in that loose sense. The companion's growth sits in L6 because it never depends on e. There is no daily score Ψ in the current code, so none is listed.



Proposition 1 (The anchor)**.** *Let* $T(e) = \mathrm{clip}(e + a(e)(h-e) + b\,s(e))$ *be master's step and* $a_{\mathrm{lo}} = \min(a_0, a_{\max}, a_{\mathrm{cap}}) > 0$*. Then* $\|T(e) - h\| \le q\|e - h\|$ *with* $q = 1 - a_{\mathrm{lo}}$ *(*$q = 0.88$ *with the defaults). Hence* h *is the unique fixed point of* T*, every orbit from every point of* $[0,1]^9$ *converges to it (the basin is the whole cube), and* $V(e) = \|e - h\|$ *is a Lyapunov function,* $V(T(e)) \le qV(e)$ *in the sense of Lyapunov (1992).*

Proof. The inequality is Proposition 2. If $T(e^*) = e^*$ then $\|e^* - h\| \le q\|e^* - h\|$ with $q < 1$, so $e^* = h$. Iterating gives $\|T^t(e) - h\| \le q^t\|e - h\| \to 0$. □

Honest limit. T is *not* a Banach contraction in the strict pairwise sense, $\|T(x) - T(y)\| \le q\|x - y\|$ for all x, y (Banach, 1922). The step size jumps to a_{cap} when the hard movers feed each other ($\lambda > 0$), so T is discontinuous there. Sampling 4000 pairs gives a pairwise ratio of up to 12.4, while the ratio toward h never exceeds 0.871. The toward-a-point form above is what uniqueness and convergence need. Numerically, the app's rounded iteration from 500 random starts lands within 0.057 of h in at most 21 steps. At the anchor, the layer coordinates are $L_1 = h = (0, 0, 0, 0, 0.8, 0.8, 0.8, 0, 0)$, $\pi_2 = 0$, $\pi_3 = (P, C, G) = (1, 0, 0)$, $\pi_4 = (0, 0, 0, 0)$ and $\pi_5 = (0, 0.8)$. With faith on, the anchor is described as rest in Christ: "This hope we have as an anchor of the soul, a hope both sure and steadfast" (Hebrews 6:19, WEB). The words change, not the math.

Time. Each mover gets a time constant τ_i (hours; Table 1). In continuous time, $\dot{e}_i = -(e_i - A_i)/\tau_i^{\mathrm{eff}} + \text{inputs}$, so over a gap dt, $e_i \mapsto A_i + (e_i - A_i)e^{-dt/\tau_i^{\mathrm{eff}}}$ with $\tau_i^{\mathrm{eff}} = \tau_i(1 + e_{\mathrm{rum}})/(1 - \rho)$. Rumination lengthens every τ, as rumination goes with longer emotions (Verduyn and Lavrijsen, 2015). The map contracts per gap with $k(dt) = \max_i e^{-dt/\tau_i^{\mathrm{eff}}} < 1$ for every $dt > 0$ and $\rho < 1$. The app shows the time to come within 0.1 per mover, $t = \max_i \tau_i^{\mathrm{eff}} \ln(|e_i - A_i|/0.1)$, only as a wide range in words, never as a promise. The popular "90-second" rule for an emotion's chemical wave is not peer-reviewed and is not used.

Resistance. Resistance $\rho \in [0, 0.9]$ is built as $\rho = \min\big(0.9,\ [1 - (1 - 0.4e_{\mathrm{avoid}})(1 - 0.5c_{\mathrm{sup}})(1 - 0.3e_{\mathrm{rum}})(1 - 0.3\,\iota)(1 - 0.3c_{\mathrm{sustain}})](1 - 0.5c_{\mathrm{will}})\big)$. Its parts:

- experiential avoidance (Hayes et al., 2006; Bond et al., 2011);
- suppression or fighting words, which tend to bring a thought back (Wegner et al., 1987);
- rumination;
- emotional inertia ι, the lag-1 autocorrelation of the saved days (Kuppens et al., 2010);
- sustain talk (Miller and Rollnick, 2013);
- willingness and acceptance words, which lower ρ.

Mover	τ range (h)	Source
fear	0.25–3	among the shortest emotions, shorter than anxiety (Verduyn and Lavrijsen, 2015)
shame	0.25–2	the shortest on average, about half an hour; guilt lasts longer (Verduyn and Lavrijsen, 2015)
anger	0.5–48	irritation is short; longer forms such as hatred last days (Verduyn and Lavrijsen, 2015)
sadness	24–120	the longest emotion, about 120 h on average to a lasting return (Verduyn and Lavrijsen, 2015)
calm, connection, hope	1–24	design choice (no duration data for the grounding movers)
rumination, avoidance	2–48	design choice; rumination and re-occurrence prolong emotions (Verduyn et al., 2009; Verduyn and Lavrijsen, 2015)

Table 1: Time constants. The typical value is the geometric mean of the range.

Resistance slows time ($\tau^{\text{eff}} = \tau/(1-\rho)$) and weakens the pull ($a_{\text{eff}} = a(1-\rho)$). The anchor step still contracts, with $q_\rho = 1 - a_{\text{lo}}(1-\rho) \le 0.995 < 1$. The signs are as intended: $\partial\rho/\partial c_{\text{sup}} > 0$, so pushing harder raises resistance, and $\partial\rho/\partial c_{\text{will}} < 0$, so willingness lowers it. The weights are modeling choices.

The moving anchor. The anchor is a moving focus point, not one fixed point. It is updated as $A_k = \text{clip}\big(\lambda A_{k-1} + (1-\lambda)(\beta e + (1-\beta)d)\big)$, with $\lambda = 0.6$ and $\beta = 0.25 + 0.25\rho$. Here d is the desired feelings (the freeing mix, else h), and $A_{-1} = h$. Within an epoch A_k is fixed, and $e \mapsto \text{clip}(e + a_{\text{eff}}(A_k - e))$ contracts toward it with q_ρ, so A_k is that epoch's unique fixed point. A transition to A_{k+1} (called "collapse" only as a picture; nothing quantum) happens when $\|e - A_k\| \le \varepsilon_{\text{in}} = 0.15$ has held for a dwell time τ_{dwell}. The dwell breaks only when $\|e - A_k\| > \varepsilon_{\text{out}} = 0.25$; this hysteresis prevents flip-flopping. With d fixed, $\|A_{k+1} - d\| \le q_A \|A_k - d\| + (1-\lambda)\beta\varepsilon_{\text{out}}$, with $q_A = \lambda + (1-\lambda)\beta \le 0.79$. So the chain stays in the cube and approaches a ball of radius $\beta\varepsilon_{\text{out}}/(1-\beta) \le 0.23$ around d geometrically. A stone is added only when A moves by at least 0.02. A sample chain (sad, replaying, $\rho = 0.40$, $d = (\dots, 0.9, 0.6, 0.7, \dots)$ then $(\dots, 0.7, 0.9, 0.8, \dots)$) runs calm/connection/hope $0.78/0.71/0.73 \to 0.81/0.68/0.72 \to \cdots \to 0.86/0.64/0.71$ by 71 h, then turns toward the new desire, reaching 0.72/0.87/0.79 by 205 h. The path is kept on the device only.

Expression governor. The goal is full expression up to a safe limit; the governor offers a gentle pause, never a block. It uses three signals:

- unawareness U: numb or detached words, flat repetition and low self-awareness;
- irritability Y: anger rising per hour, and irritation words;
- mania-like activation M_n: racing words, pressured long entries, true size G, short-sleep words, and bursts of entries.

From these it forms the activation $\alpha = 1 - (1-Y)(1-M_n)(1-\frac{1}{2}U)$ and a limit $\ell(C) = 0.45(1 - 0.5C)$, which tightens as conflict rises. A PI controller with leak and anti-windup (Åström and Murray, 2008) runs on the error $E = \alpha - \ell(C)$: $I_k = \text{clip}_{[-1,1]}(0.8 I_{k-1} + E_k)$, frozen while u is saturated, and $u_k = \text{clip}_{[0,1]}(1.2E_k + 0.4I_k)$. The output is graded: keep writing; a slow-down pause; breathe and ground (STOP; Linehan, 2015; an early pause in the sense of Gross, 1998); step away. *Stability (honest scope):* u and I are bounded by the clips. The leak makes the controller state a contraction, $|I_k - I'_k| \le 0.8|I_{k-1} - I'_{k-1}|$, so old error fades geometrically. The governor never feeds back into e, so q, q_A and q_ρ are untouched. The loop through the person is not modeled. If racing words or short sleep appear on three or more days, a gentle line suggests talking with a doctor. This is not a diagnosis; DSM-5 (American Psychiatric Association, 2013) is context only. Crisis wording still shows only the 988/911 box.

Scenes and About me. Thirteen small original SVG scenes follow the parable's beats with the same traveler and fawn. The text and the pictures come from one list of beats, so they stay in sync. A grounding picture is chosen from the governor (activation or true-size drift: a shared table or a garden; irritability: open water or a horizon; unawareness: hands or a lamp) and is paired with one kind, concrete action. This draws on positive imagery (Holmes and Mathews, 2010), nature and attention restoration (Kaplan, 1995; Berman et al., 2008) and kindness and well-being, a small average effect (Curry et al., 2018). The optional, local-only *About me* (hobbies, field) does four things: it adds a behavioral-activation step (Lewinsohn, 1974; Martell et al., 2001), a field motif

and metaphor, one more song or film, and a word of room for flow (Csikszentmihalyi, 1990). It never changes a number.

6 What is proven, what is tested, what is modeled

Proposition 2 (Contraction still holds)**.** *Let T be the step map (2) with hard and process homes at 0. Then for every $e \in [0,1]^9$, $\|T(e) - h\| \le (1 - a_{\text{lo}})\|e - h\|$ with $a_{\text{lo}} = \min(a_0, a_{\max}, a_{\text{cap}}) > 0$. The Kai term does not change T, so the iteration converges to $e^* = h$, where $P = 1$.*

Proof. For a grounding mover, $T(e)_j - h_j = (1-a)(e_j - h_j)$. For a hard or process mover, $T(e)_j = \text{clip}_{[0,1]}(e_j(1 - a - b\kappa_j))$ with $\kappa_j = -\sum_g \min(0, W_{gj})h_g \ge 0$, so $0 \le T(e)_j \le (1-a)e_j$. Summing squares gives the bound. The Kai term enters only line (1): it takes a point of $[0,1]^9$ to another point of $[0,1]^9$ (clipped), raising only anger, shame and fear by at most Xw_j. The proof needs nothing else from the starting point, and the solve and recovery iterate T with empty text, where $X = 0$. □

Remark (If the pressure were applied at every step). Suppose the map were instead $T_X(e) = \text{clip}_{[0,1]}(T(e) + p)$ with a constant $p = X(1 - S)w \ge 0$ on the hard movers. Then $|T_X(e)_j - h_j| \le (1-a)|e_j - h_j| + p_j$, so $d_{t+1} \le (1 - a_{\text{lo}})d_t + \|p\|$ and $\limsup d_t \le \|p\|/a_{\text{lo}}$. That is bounded, but it does *not* converge to h. This bound is loose (with the defaults $a_{\text{lo}} = 0.12$). The app does not do this: outside hurt is applied once per reading.

Proposition 3 (The shield behaves as the story says)**.** *$S \in [0, 0.99]$ and $\partial S/\partial V = \prod_{U \ne V}(1 - U) \ge 0$ for each virtue V. With X fixed, $X(1-S)$ and the pushed anger fall as S rises, so u and $I = X(1-S)u$ are non-increasing in S; I is strictly decreasing when $X > 0$ and $u > 0$ (before rounding). With $X = 0$ nothing changes: no push, $I = 0$, and no card tile.*

Tested, not proven: the code matches (1)–(3) to the cent. The rounded iteration stops within $0.005/a$ of h. The detection separates hurt received from hurt given on the listed examples. Crisis wording returns only the 988/911 box. Everything is deterministic. The unit suite has 299 checks. **Modeled, not validated:** the movers' links, every weight and threshold, the cue lists, the virtue formulas, and the practice rules. No study with people has been run.

Credits and References

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Every source used across MindFrame (the app, its README and its papers) was checked in October 2026. Journal articles were checked against Crossref (title, authors, year, volume, pages, DOI); books against Open Library or publisher and library catalogs; reports, preprints, archives and web articles at their source. 111 are verified; 2 are flagged and marked below. Films are credited to their studios and filmmakers and songs to their artists and songwriters; MindFrame only names them and quotes no lyrics. The works cited in this paper follow under *Works cited*.

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